

ADR-003: Three-Layer Experimental Architecture and Phased Governance Design

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Supersedes / relates to: Builds on ADR-001 and ADR-002. Defers the bespoke bridge application considered during design discussions to future work.

1. Context

The Dossier Space pilot stress-tests out-of-the-box agentic AI against real evidentiary complexity and volume, using the Enron corpus (CMU set for prototyping; EDRM v2, ~73GB, for validated local testing). A design question arose: should the pilot use Claude Code as supplied, or should a bespoke "bridge" application mediate between the AI and the evidence store, providing chat, tool execution, and custody logging under our own code?

The question matters because the SCL paper’s central argument concerns evidentiary custody and data sovereignty. A mediation layer that hashes, timestamps, and chain-logs every access would generate an infrastructure-level custody record independent of the model’s self-report — a materially stronger evidential claim than a transcript alone. But a bespoke bridge would also reimplement the agentic loop (context management, retrieval strategy, retries), meaning every observed capability or failure would be partly attributable to our engineering rather than to the AI under test.

2. Decision

We adopt a three-layer architecture that separates the research subject, the experimental environment, and the research method, and we execute the pilot as a phased design in which governance is treated as an independent experimental variable. No bespoke bridge application is built for the research phase. Claude Code is used unmodified as the experimental subject; a deliberately minimal MCP server provides the evidence store and custody machinery; primers and ROMER are applied only as declared treatments in later phases.

2.1 The three layers

Layer	Contents	Rule
1. Research subject	Claude Code (or another out-of-the-box agentic AI client and its underlying model)	Never modified, extended, or compensated for. Version-pinned and recorded per run.
2. Experimental environment	MCP server, evidence store, custody log, permissions and scoping, ROMER capture points	Contains policy, not intelligence. No summarising, ranking, or inference. Frozen across phases.
3. Research method	Primers, task definitions, metrics, evaluation protocol, reporting	Defined before execution; varied only as declared experimental treatments (Phases 3a/3b).

The governing principle for Layer 2 is that the mediation layer contains policy, not intelligence. It acts as a gatekeeper and recorder — checking each request against scope rules, executing it, hashing and chain-logging the result — and never as a co-reasoner. The MCP server is the laboratory bench: it provides controlled conditions; it does not perform the chemistry.

2.2 MCP server tool set (Layer 2)

The server exposes a small, fixed set of non-inferential tools: `read_file`, `list_folder`, `search_messages`, `get_headers`, `hash_document`, and `verify_hash`. All tools are read-only at server level; scoping restricts access to defined custodians, folders, or date ranges; a privilege-gate mechanism refuses to serve designated material regardless of the request. Every tool response carries the hash of the source artefact it derives from, and every request–response pair is written to a tamper-evident log in which each entry incorporates the hash of its predecessor.

Tool descriptions are treated as experimental material, not documentation. In MCP, descriptions are transmitted to the model and shape its behaviour; they are therefore frozen for the duration of the experiment, recorded verbatim in the method, and published with the results. Any change to a tool description constitutes a change of treatment and requires a new experimental run.

2.3 Phase structure

Phase	Configuration	What changes relative to prior phase	What the delta measures
1	Claude Code → direct filesystem → corpus	Nothing (baseline)	Out-of-the-box agentic capability
2a	Claude Code → MCP passthrough (custody off) → corpus	Mediation protocol only	Cost of indirection
2b	Claude Code → MCP with custody (hashing, chained logging, scoping) → corpus	Custody machinery only	Cost of governance, isolated
3a	Phase 2b + primer (CIPDA route)	Structural prompting only	Effect of methodological structure
3b	Phase 3a + ROMER reasoning traces	Reasoning trace requirement only	Effect of trace discipline; trace-vs-log gap becomes measurable

Phase 2 is split into a null-mediation control (2a) and the custody configuration (2b) to remove a confound: moving from direct filesystem access to MCP changes both the mediation protocol and the custody machinery at once. The 2a–2b delta isolates governance overhead specifically — the number the paper reports as the cost of evidentiary governance. Phase 3 is likewise split so that the primer’s contribution (3a) is separable from ROMER’s (3b); reporting them as a single combined stack would leave reviewers unable to attribute any improvement.

2.4 Experimental discipline

The subject will not sit still: Claude Code updates frequently and the underlying model changes. Client and model versions are pinned where the tooling permits and recorded in every ROMER trace and custody log entry regardless. Phases are run interleaved rather than strictly sequentially, so that subject drift manifests as noise rather than systematic bias between phases. Because agentic runs are non-deterministic, each task is repeated multiple times per phase; distributional spread is reported alongside means, consistent with the variance-suppression framing established in the CIPDA N=25 study — the hypothesis being that governance layers may narrow variance even where they do not improve central performance.

3. Rationale — how we arrived here

The decision record deliberately preserves the reasoning path, because the rejected alternatives clarify what the accepted design is for.

The starting proposal was a bespoke bridge application: a purpose-built Windows client mediating between any AI provider and the evidence store, with custody logging built in. Its attraction was total control of the custody layer and provider independence — every byte passing through code we wrote. Its fatal defect for this pilot was methodological: the research question is "how far can out-of-the-box AI go with real evidence complexity and volume," and a bespoke bridge changes the experimental object. Capabilities become partly our engineering; failures become ambiguous between the AI and the app. The bridge would also have required months of reimplementing an agentic loop that Claude Code already supplies, in order to produce a result that tested our bridge rather than the AI.

The second position was a hybrid: Claude Code unmodified as client, plus a thin MCP custody server — governance without contaminating the subject. This survives into the final design, but two refinements proved necessary. First, the three-layer formalisation (proposed by Leonard) makes the separation explicit and auditable: the AI appears in Layer 1 only, and disappears entirely from the environment and the method. Second, the phase structure was hardened against confounds: the null-mediation control (2a) separates indirection cost from governance cost, and the 3a/3b split separates the primer's effect from ROMER's.

The reframing that settles the design is this: the pilot's contribution is not "we used MCP," which many will do, nor a claim that any one tool posture is superior. It is the demonstration that AI capability and evidential governance are separable properties that can be varied and measured independently — governance as an engineered, quantifiable experimental variable rather than a qualitative aspiration. That claim requires exactly this architecture: an untouched subject, a policy-only environment, and treatments introduced one at a time. Anything richer than the minimal MCP server weakens the claim; anything less than the custody machinery forfeits the evidential story.

A secondary rationale is evidential: the architecture yields two independent records of every run — the model's ROMER self-report and the server's infrastructure log of what was actually touched. The gap between them is itself a measurable finding, and one a court or opposing expert would regard as material. A bespoke bridge could also have produced this, but only the minimal-server design produces it without compromising the experiment.

4. Consequences

Positive: the experimental object stays clean and the paper's central claim — governance as an independent variable — becomes defensible against methodological challenge. The MCP server (Dossier Space MCP Server v1) becomes a citable, versioned artefact and the custody log a tamper-evident evidential record independent of model self-report. The design plays directly to prior work: variance analysis extends the CIPDA study, and the trace-versus-log gap operationalises ROMER's assurance claim. Infrastructure friction encountered while imposing this discipline remains reportable as a field finding, consistent with the pilot's method to date.

Negative and accepted: the phased, repeated, interleaved design multiplies run count and elapsed time. Version pinning may be imperfect where tooling does not expose it; this residual drift is recorded rather than eliminated. Corpus content read by the model still travels to the provider's API for inference — the architecture governs access and custody records, not data locality. The honest description of this posture is "workstation-mediated with logged custody, cloud inference"; fully sovereign inference via a local model is compatible with the same Layer 2 and is deferred, alongside the bespoke bridge, to production-oriented future work.

Deferred: the bespoke bridge application is repositioned as a production deliverable, to be specified after the pilot shows where stock tooling breaks. Nothing in this ADR forecloses it; the three-layer separation means Layer 2 carries forward into any future client unchanged.

5. Actions

Specify and freeze the Layer 2 tool set and tool descriptions; design the chained custody log schema; implement Dossier Space MCP Server v1 in Python against the CMU corpus; define the task battery, metrics, and repetition counts (Layer 3) before any measured run; record client and model versions in all logs and traces; execute phases interleaved; report means with distributional spread.